

CUSTOMER SEGMENTATION REPORT

Task 6: Customer Segmentation

Dataset: Mall Customer Dataset

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1. Introduction

Customer segmentation is an important business analytics technique used to group customers with similar characteristics and purchasing behavior. This project uses the Mall Customer Dataset and K-Means Clustering to identify distinct customer groups.

2. Task Details

Objective: Group customers based on behavior. Required tasks included data preprocessing, applying K-Means clustering, visualizing clusters, and generating customer insights.

3. Dataset Overview

The Mall Customer Dataset contains 200 customer records and 5 attributes: Customer ID, Gender, Age, Annual Income (k\$), and Spending Score (1-100).

4. Data Preprocessing

The dataset was inspected for quality issues and prepared for clustering. Annual Income and Spending Score were selected as key features. Feature scaling was applied using StandardScaler to ensure equal contribution of variables during clustering.

5. K-Means Clustering

The Elbow Method was used to determine the optimal number of clusters. Based on the WCSS trend, 5 clusters were selected. K-Means clustering was then applied to group customers with similar income and spending behavior.

6. Cluster Visualization

A scatter plot was created using Annual Income and Spending Score, with customers colored according to their assigned cluster. This visualization clearly highlighted distinct customer segments.

7. Customer Segment Insights

Cluster 1: High income and high spending customers (premium customers).
Cluster 2: High income and low spending customers (potential growth segment).
Cluster 3: Low income and high spending customers.
Cluster 4: Low income and low spending customers.
Cluster 5: Average income and average spending customers.

These segments can help businesses design targeted marketing strategies and improve customer engagement.

8. Business Recommendations

Businesses should focus on retaining premium customers, create personalized offers for high-income low-spending customers, and develop targeted campaigns for each customer segment to maximize revenue and customer satisfaction.

9. Conclusion

The project successfully grouped customers into meaningful segments using K-Means clustering. The identified customer groups provide valuable insights that can support marketing, sales, and customer relationship management decisions.